

Mihir Balsara

617-642-4806 | mihirnbalsara@gmail.com | [linkedin.com/in/mbalsara](https://www.linkedin.com/in/mbalsara) | github.com/GlxryX

EDUCATION

Georgia Institute of Technology

Bachelor of Science in Computer Science - GPA: 3.73

Atlanta, GA

Expected Dec. 2026

EXPERIENCE

Software Development Intern

PowerPlan

June 2025 – Present

Atlanta, GA

- Designed and maintained database schemas and wrote complex SQL queries for a distributed, cloud-native application serving 10,000+ users.
- Implemented scalable microservices architecture using Docker containerization and Kubernetes orchestration.
- Collaborated in an Agile/Scrum environment, utilizing CI/CD pipelines to deploy and test new features.

Cybersecurity Research Intern

Georgia Tech Research Institute

May 2024 – May 2025

Atlanta, GA

- Collaborated with an international team (SSLab, GTRI, KAIST) for the DARPA AIxCC Finals. Qualified for DEFCON 32.
- Developed high-performance testing and data mutation frameworks in Python and C, applying knowledge of computer architecture and algorithms.
- Implemented and managed CI/CD pipelines with Jenkins, automating builds and reducing deployment time by 40%.

Research Assistant

Lincoln Labs

Aug. 2022 – Jul. 2023

Lexington, MA

- Applied data mining and statistical modeling to 500k+ data points, using Python to generate actionable business insights.
- Built automated data visualization pipelines with Matplotlib/Seaborn, reducing report generation time by 60%.
- Authored and published peer-reviewed findings, demonstrating strong analytical and communication skills.

Data Engineer

Russian School of Mathematics

Jun. 2022 – Aug. 2022

Lexington, MA

- Engineered a Big Data ETL pipeline in Python to process and model 2M+ student records, improving efficiency by 30%.
- Collaborated with cross-functional teams to implement robust data validation and error handling mechanisms.

PROJECTS

Five-Stage Pipelined RISC CPU | C, Assembly, Digital Logic, Processor Design, Hazard Resolution

- Implemented a five-stage pipelined RISC CPU in CircuitSim, cutting execution latency through parallelization, a custom forwarding unit, and single-bubble load hazard resolution.
- Achieved a 4.6x latency reduction over non-pipelined baseline.

Spotify Wrapped Mobile App | Java, Firebase, Machine Learning, REST APIs

- Integrated web services (Spotify REST API) and applied machine learning algorithms to generate personalized music recommendations.
- Developed a cross-platform mobile application with Firebase authentication and real-time data synchronization.
- Implemented a comprehensive unit testing suite with JUnit, achieving 90%+ code coverage for core functionality.

1Record - Cornell Health RX Winner | Python, AI, NLP, MongoDB, React

- Implemented an AI system using Natural Language Processing (LLM) to derive predictive insights from EHR data.
- Developed a full-stack healthcare application using React/TypeScript frontend with a Python backend.
- Collaborated in a 48-hour hackathon environment, demonstrating rapid prototyping and problem-solving skills.

Discord Applications Platform | JavaScript, Node.js, Distributed Systems, Docker

- Architected and deployed a scalable distributed system serving 346,075+ users with 99.9% uptime.
- Optimized application performance through profiling and benchmarking, reducing response times by 35%.
- Built a CI/CD pipeline with Jenkins, reducing deployment downtime to under 5 minutes and enabling rapid iterations.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, SQL, C#, JavaScript, TypeScript, Go

Core Competencies: Distributed Systems, Algorithms, Data Structures, Cloud Computing, Object-Oriented Design, Database Systems, Machine Learning, AI, Big Data, Operating Systems (Linux)

Tools & Platforms: AWS, Docker, Kubernetes, Git, Jenkins, CI/CD, Agile/Scrum, Jest, JUnit

Databases: MongoDB, MySQL, Firebase, Postgres, Oracle DB